THE MIRROR-PAIRED HADAMARD SQUARE-DENOMINATOR IDENTITY, THE UNIVERSAL CONSERVATION LAW $\int_{\mathbb{R}} K\,\mathrm{d}u \equiv \pi$, AND STATIONARY TRUDGIAN VARIATION IN THE CRITICAL STRIP

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Abstract. We establish an exact dual representation linking the normalized transverse Wronskian $\widehat{W}(d,t) = W(d,t)/|\Xi(t-\mathrm{i}d)|^2$ of Riemann's Ξ-function to Fejér–Lukács pure-square trigonometric multiplets $\mathcal{S}_\pm(d,t)$suppressing the classical de la Vallée Poussin $\sigma = 1$ ($d = \frac{1}{2}$) pole barrierand to mirror-paired Hadamard zero sums on the spectral side. Pairing each nontrivial zero $\rho = \frac{1}{2} + d_\rho + \mathrm{i}\gamma_\rho$ with its reflection $1-\bar{\rho} = \frac{1}{2} - d_\rho + \mathrm{i}\gamma_\rho$ yields the square-denominator kernel $K(d, d_\rho, u) = d(d^2 - d_\rho^2 + u^2)/[((d^2 - d_\rho^2 + u^2)^2 + 4u^2 d_\rho^2)]$, which satisfies the universal Guinand–Weil/Hadamard conservation law $\int_{-\infty}^{\infty} K(d, d_\rho, u)\,\mathrm{d}u \equiv \pi$ for all $d > |d_\rho|$ and admits on $d \geq \frac{1}{2}$ the hypothesis-free minorant $K_{\min}(d,u) := K(d, \frac{1}{2}, u)$ throughout $|u| \geq H_{\min}(d) = \sqrt{(d^2 - 1/4)/3}$. Integrating by parts against $\mathrm{d}N(\gamma) = \frac{1}{2\pi}\ln(\gamma/2\pi)\,\mathrm{d}\gamma + \mathrm{d}S(\gamma)$ with Trudgian's bound $|S(T)| \leq B_S(T)$ factors $\partial_H K_{\min}(d,H) < 0$ out of $\partial_H \mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}(d, kt; H) = 0$, pinning the optimal window half-width to the d-independent stationary trajectory $H_{\mathrm{opt}}(kt) \equiv 2\pi B_S(kt)/\ln(kt/2\pi) \to 2\pi(0.112)$ and contracting the multi-harmonic strip boundary across $t \in [14, 10^8]$.

1. Introduction and Main Results

In his foundational 1859 memoir, Riemann [21] introduced the completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and its critical-line rotation $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z)$, which is entire of order 1 and real-valued on $\mathbb{R}$ [7, 20, 26]. Decomposing $\Xi(t - \mathrm{i}d) = R(d,t) + \mathrm{i}\,I(d,t)$ for spectral height $t > 0$ and transverse distance $d = \sigma - \frac{1}{2} > 0$, the transverse Wronskian $W(d,t) := R(d,t)I_t(d,t) - I(d,t)R_t(d,t) = \frac{1}{2}\partial_d(|\Xi(t-\mathrm{i}d)|^2)$ and its scale-invariant normalization $\widehat{W}(d,t) := W(d,t)/|\Xi(t-\mathrm{i}d)|^2 = \partial_t \arg \Xi(t - \mathrm{i}d) = \mathrm{Re}(\xi'(\frac{1}{2}+d+\mathrm{i}t)/\xi(\frac{1}{2}+d+\mathrm{i}t))$ govern the vertical monotonicity of $|\Xi(t-\mathrm{i}d)|^2$ and the zero-detection criteria of Speiser [24], Hinkkanen [9], and Lagarias [11].

In Papers I–VI of this series [13, 14, 15, 16, 17, 18], the Wronskian $W(d,t)$ was formulated in Wigner phase-space coordinates $(x,y) = (u+v, u-v)$ [29, 10] and analyzed via complex saddle-point coalescence, parity-reduced moment recurrences, Chladni nodal chamber topology, de Branges–Jensen secant kernels [2], universal mother-wavelet dilation [23, 5, 19], and the sieved mirror kernel $\Phi_X(u)$ associated with finite-divisor Euler-product gauges [22, 12, 1]. A central analytical problem arises at the interface between the arithmetic Dirichlet-series representation of $\widehat{W}(d,t)$ and its spectral Hadamard partial-fraction representation [8, 3]: in the classical trigonometric method of de la Vallée Poussin, the inequality $3 + 4\cos\theta + \cos 2\theta = 2(1+\cos\theta)^2 \geq 0$ converts the logarithmic derivative $-\mathrm{Re}(\zeta'/\zeta)(\frac{1}{2} + d + \mathrm{i}kt)$ into a non-negative pure-square prime sum in the Euler-product half-plane $d > \frac{1}{2}$ ($\sigma > 1$), but must contend with the pole of $-\zeta'(\frac{1}{2}+d)/\zeta(\frac{1}{2}+d)$ as $d \to (\frac{1}{2})^+$ and

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with the background Hadamard zero sum when inferring zero-free regions inside the critical strip $0 < d < \frac{1}{2}$.

The present paper (Paper VII) resolves this interface by coupling Fejér–Lukács pure-square trigonometric multiplets [4, 25] with the mirror-paired Hadamard square-denominator identity, a finite-cutoff sieved pure-square decomposition, an exact π-conservation law, a hypothesis-free far-field minorant $K_{\min}(d,u)$, and a d-independent stationary variational principle driven by Trudgian's explicit bound on $S(T)$ [27]. Our main results are consolidated in the following theorem.

Theorem 1.1 (Mirror-Paired Hadamard Square-Denominator Identity, Universal Conservation Law, and Stationary Trudgian Variation)**.** *Let $s = \frac{1}{2} + d + \mathrm{i}t$ with $d > 0$ and $t > 0$, and let $\widehat{W}(d,t) = \mathrm{Re}(\xi'(s)/\xi(s)) = \sum_\rho \frac{d-d_\rho}{(d-d_\rho)^2+(t-\gamma_\rho)^2}$ be the Hadamard partial-fraction expansion over the nontrivial zeros $\rho = \frac{1}{2} + d_\rho + \mathrm{i}\gamma_\rho$ of $\zeta(s)$. Then the following four structural laws hold:*

(i) ***Pure-Square Trigonometric Multiplets and Fejér–Lukács Pole Suppression.*** *For any non-negative trigonometric polynomial $T_K(\theta) := \left|\sum_{j=0}^K a_j \mathrm{e}^{\mathrm{i}j\theta}\right|^2 = c_0 + \sum_{k=1}^K c_k \cos(k\theta) \geq 0$ with $c_0 = \sum_{j=0}^K a_j^2 > 0$ and $c_k = 2\sum_{j=0}^{K-k} a_j a_{j+k} \geq 0$, define the dual pure-square prime sums on $d > \frac{1}{2}$ by $\mathcal{S}_\pm(d,t) := \sum_{n=2}^\infty \Lambda(n) n^{-(1/2+d)} T_K(t\ln n + \frac{1\mp 1}{2}\pi) \geq 0$. Then $\widehat{W}(d,kt)$ satisfies the exact dual identities*

$$\begin{aligned} \sum_{k=1}^K c_k\, \widehat{W}(d,kt) + \mathcal{S}_+(d,t) &\equiv \sum_{k=1}^K c_k\, \mathcal{A}(d,kt) + c_0 M(d), \\ \sum_{k=1}^K (-1)^{k-1} c_k\, \widehat{W}(d,kt) - \mathcal{S}_-(d,t) &\equiv \sum_{k=1}^K (-1)^{k-1} c_k\, \mathcal{A}(d,kt) - c_0 M(d), \end{aligned} \tag{1.1}$$

where $\mathcal{A}(d,t) := \mathcal{A}_\Gamma(d,t) + \mathcal{P}_{\mathrm{pole}}(d,t)$ is the Archimedian carrier and $M(d) := -\zeta'(\frac{1}{2}+d)/\zeta(\frac{1}{2}+d) > 0$. Over all real amplitude vectors $(a_0,\dots,a_K) \neq 0$, the first-harmonic ratio c_1/c_0 is bounded above by the exact Toeplitz Rayleigh maximum $\Lambda_{\max}(K) = 2\cos(\frac{\pi}{K+2})$, attained uniquely by the Fejér–Lukács sine eigenvector $a_j^{\mathrm{FL}} = \sin(\frac{(j+1)\pi}{K+2})$ $(0 \leq j \leq K)$, which achieves the closed-form $\sigma = 1$ pole-suppression ratio $(\sum_{k=1}^K c_k^{\mathrm{FL}})/c_0^{\mathrm{FL}} = \frac{2}{K+2}\cot^2(\frac{\pi}{2(K+2)}) - 1$.

(ii) ***Mirror-Paired Hadamard Square-Denominator Identity and Sieved Strip Penetration.*** *Pairing each zero $\rho = \frac{1}{2} + d_\rho + \mathrm{i}\gamma_\rho$ with its functional-equation mirror $1 - \bar{\rho} = \frac{1}{2} - d_\rho + \mathrm{i}\gamma_\rho$ and setting $u := t - \gamma_\rho$ yields the exact kernel identity*

$$K(d,d_\rho,u) := \frac{1}{2}\left[\frac{d-d_\rho}{(d-d_\rho)^2+u^2} + \frac{d+d_\rho}{(d+d_\rho)^2+u^2}\right] \equiv \frac{d\,(d^2-d_\rho^2+u^2)}{\left(d^2-d_\rho^2+u^2\right)^2 + 4u^2 d_\rho^2}, \tag{1.2}$$

whose denominator $D(d,d_\rho,u) \equiv (d^2-d_\rho^2+u^2)^2 + 4u^2d_\rho^2 \equiv (d^2+d_\rho^2+u^2)^2 - 4d^2d_\rho^2$ is strictly positive for all $u \neq 0$, while $K(d,d_\rho,u) > 0$ for all $d > |d_\rho|$ and all $u \in \mathbb{R}$. Subtracting the finite pure-square prime sum $\Delta B_X(d,t) := \sum_{n=2}^{\lfloor X \rfloor} \Lambda(n) n^{-(1/2+d)} T_K(t\ln n) \geq 0$ leaves the non-negative tail $\mathcal{S}_{+,>X}(d,t) \geq 0$ and reduces the upper budget to $U_{T,X}(d,t) := \sum_{k=1}^K c_k \mathcal{A}(d,kt) + c_0 M(d) - \Delta B_X(d,t)$, pushing the strip threshold

$$d_{\mathrm{strip},X}^{\mathrm{mirror}}(t) := \inf_{d>1/2} \sqrt{\max\left(0,\, d^2 - \frac{2c_1 d}{U_{T,X}(d,t)}\right)}$$

monotonically down to 0.000000 at $t \in \{\gamma_1,\gamma_2,\gamma_3,60,100\}$ for $X = 1000$.

(iii) ***Universal Conservation Law $\int_{\mathbb{R}} K\,\mathrm{d}u \equiv \pi$ and Hypothesis-Free Minorant $K_{\min}(d,u)$.*** *For every $d > |d_\rho|$ with $d_\rho \in (-\frac{1}{2},\frac{1}{2})$, the mirror-paired Hadamard kernel satisfies the exact*

conservation law and tail fraction identity

$$\int_{-\infty}^{\infty} K(d,d_\rho,u)\,\mathrm{d}u \equiv \pi, \quad \alpha_{\mathrm{tail}}(d,H) = 1 - \frac{1}{\pi}\left[\arctan\frac{H}{d-\frac{1}{2}} + \arctan\frac{H}{d+\frac{1}{2}}\right], \tag{1.3}$$

together with the exact algebraic difference factorization (with $W := d^2+u^2$)

$$K(d,d_\rho,u) - K\Big(d,\frac{1}{2},u\Big) \equiv \frac{d\,(\frac{1}{4}-d_\rho^2)\,[3u^4+(2d^2+d_\rho^2+\frac{1}{4})u^2-(d^2-\frac{1}{4})(d^2-d_\rho^2)]}{((d^2-d_\rho^2+u^2)^2+4u^2d_\rho^2)\,((d^2-\frac{1}{4}+u^2)^2+u^2)}. \tag{1.4}$$

Consequently, for all $d \geq \frac{1}{2}$, the boundary kernel $K_{\min}(d,u) := K(d,\frac{1}{2},u) = \frac{d(d^2-1/4+u^2)}{(d^2-1/4+u^2)^2+u^2}$ is a strictly positive, hypothesis-free minorant $0 < K_{\min}(d,u) \leq K(d,d_\rho,u)$ for all $|d_\rho| < \frac{1}{2}$ and all $|u| \geq H_{\min}(d) := \sqrt{(d^2-1/4)/3}$ $(u \neq 0)$.

(iv) ***The d-Independent Stationary Variational Law*** $H_{\mathrm{opt}}(kt) \equiv \frac{2\pi B_S(kt)}{\ln(kt/2\pi)}$. *Under Stieltjes integration by parts of $\mathrm{d}N(\gamma) = \frac{1}{2\pi}\ln\frac{\gamma}{2\pi}\,\mathrm{d}\gamma + \mathrm{d}S(\gamma)$ with Trudgian's explicit bound $|S(T)| \leq B_S(T) := A_S\ln T + B_S^{(0)}\ln\ln T + C_S$, the plateau background lower bound $\mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}(d,kt;H) := \frac{1}{2\pi}\ln\frac{kt}{2\pi}\,\Phi_{\mathrm{plat}}(d,H) - 2B_S(kt)K_{\min}(d,H)$ with $\Phi_{\mathrm{plat}}(d,H) := \pi\alpha_{\mathrm{tail}}(d,H) + 2HK_{\min}(d,H)$ obeys the exact derivative factorization*

$$\frac{\partial\mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}}{\partial H}(d,kt;H) \equiv 2\,\frac{\partial K_{\min}}{\partial H}(d,H)\left[\frac{H}{2\pi}\ln\frac{kt}{2\pi} - B_S(kt)\right] = 0 \quad\Longleftrightarrow\quad H_{\mathrm{opt}}(kt) \equiv \frac{2\pi\,B_S(kt)}{\ln(kt/2\pi)}, \tag{1.5}$$

at which the boundary penalty vanishes identically, $\mathcal{B}_{\mathrm{bg}}^{\mathrm{opt}}(d,kt) \equiv \frac{1}{2}\alpha_{\mathrm{tail}}(d,H_{\mathrm{opt}}(kt))\ln(kt/2\pi)$, and $H_{\mathrm{opt}}(kt) \to H_\infty := 2\pi A_S = 0.7037167544\ldots$ as $kt\to\infty$, contracting the multi-harmonic strip boundary $d_{\mathrm{strip},2000}^{\mathrm{opt}}(t)$ across $t \in [14,10^8]$.

The canonical analytical constants governing Theorem 1.1 are given explicitly by

$$\Lambda_{\max}(K) = 2\cos\Big(\frac{\pi}{K+2}\Big), \qquad \frac{1}{c_0^{\mathrm{FL}}}\sum_{k=1}^{K} c_k^{\mathrm{FL}} = \frac{2}{K+2}\cot^2\Big(\frac{\pi}{2(K+2)}\Big) - 1,$$
$$\int_{-\infty}^{\infty} K(d,d_\rho,u)\,\mathrm{d}u \equiv \pi, \qquad (A_S,B_S^{(0)},C_S^{(0)},C_S) = (0.112,\,0.278,\,0.206,\,2.510),$$
$$H_{\min}(d) = \sqrt{\max\Big(0,\frac{d^2-1/4}{3}\Big)}, \qquad H_\infty = 2\pi A_S = 0.70371675\ldots. \tag{1.6}$$

1.1. **Proof overview.** The proof of Theorem 1.1 proceeds in three analytical stages:

(1) *Logarithmic-derivative coupling and Toeplitz Rayleigh maximization (Sections 2 and 3).* We decompose $\widehat{W}(d,t) = \mathcal{A}(d,t) - \mathcal{Z}(d,t)$ into the Archimedian carrier $\mathcal{A}(d,t)$ and the Dirichlet series $\mathcal{Z}(d,t) = -\mathrm{Re}(\zeta'/\zeta)(\frac{1}{2}+d+\mathrm{i}t)$. Forming the trigonometric combination $\sum_{k=1}^{K} c_k\widehat{W}(d,kt)$ for $T_K(\theta) = |\sum_{j=0}^{K} a_j\mathrm{e}^{\mathrm{i}j\theta}|^2 \geq 0$, we diagonalize the tridiagonal Toeplitz form $\sum_{|j-m|=1} a_ja_m$ via the sine eigenvector $a_j^{\mathrm{FL}} = \sin(\frac{(j+1)\pi}{K+2})$ to establish $\Lambda_{\max}(K)$ and the cotangent identity (1.6). We then pair ρ with $1-\bar{\rho}$ in Hadamard's partial-fraction expansion to obtain the square-denominator identity (1.2) and subtract the finite pure-square prime sum $\Delta B_X(d,t)$ to contract $d_{\mathrm{strip},X}^{\mathrm{mirror}}(t)$.

(2) *Exact contour integration and partial-fraction difference factorization (Section 4).* By absolute convergence of $\int_{\mathbb{R}}[(d\pm d_\rho)^2+u^2]^{-1}\,\mathrm{d}u$ for $d > |d_\rho|$, we evaluate the antiderivative on $[-H,H]$ and establish the universal conservation law $\int_{\mathbb{R}} K(d,d_\rho,u)\,\mathrm{d}u \equiv \pi$. Writing $W = d^2+u^2$, we place $K(d,d_\rho,u) - K(d,\frac{1}{2},u)$ over its positive common denominator and factor the numerator

as $d(\frac{1}{4}-d_\rho^2)[4u^2W-(W-d_\rho^2)(W-\frac{1}{4})]$, proving that $K_{\min}(d,u)=K(d,\frac{1}{2},u)$ minorizes every zero pair with $|d_\rho|<\frac{1}{2}$ for all $|u|\geq H_{\min}(d)$.

(3) *Stieltjes integration by parts and stationary boundary-jump cancellation (Section 5).* Decomposing $\mathrm{d}N(\gamma)=\frac{1}{2\pi}\ln(\gamma/2\pi)\,\mathrm{d}\gamma+\mathrm{d}S(\gamma)$, we replace $K_{\min}(d,u)$ on $[-H,H]$ by the flat plateau $K_{\min}(d,H)$ so that the composite profile is continuous across $|u|=H$ and vanishes in derivative on $(-H,H)$. Integrating $\int_{|u|\geq H}K_{\min}(d,u)\,\mathrm{d}S(\gamma)$ by parts against $|S(\gamma)|\leq B_S(kt)$ produces the boundary penalty $-2B_S(kt)K_{\min}(d,H)$. Because the fundamental theorem of calculus on $\pi\alpha_{\mathrm{tail}}(d,H)$ cancels the $+2K_{\min}(d,H)$ product-rule jump of $2HK_{\min}(d,H)$, we obtain $\partial_H\Phi_{\mathrm{plat}}(d,H)\equiv 2H\,\partial_H K_{\min}(d,H)$, so $2\,\partial_H K_{\min}(d,H)\neq 0$ factors out of $\partial_H\mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}(d,kt;H)=0$ and pins $H_{\mathrm{opt}}(kt)$ to (1.5).

1.2. **Notation and conventions.** Throughout this paper, $s=\sigma+\mathrm{i}t=\frac{1}{2}+d+\mathrm{i}t$ denotes a complex variable with transverse displacement $d=\sigma-\frac{1}{2}>0$ and spectral height $t>0$, related to $z=t-\mathrm{i}d$ by $s=\frac{1}{2}+\mathrm{i}z$. The completed zeta function is $\xi(s)=\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and Riemann's Ξ-function is $\Xi(z)=\xi(\frac{1}{2}+\mathrm{i}z)=R(d,t)+\mathrm{i}\,I(d,t)$. At any point (d,t) where $\Xi(t-\mathrm{i}d)\neq 0$, the Cauchy–Riemann equations $R_d=I_t$ and $I_d=-R_t$ imply that the normalized transverse Wronskian is

$$\widehat{W}(d,t):=\frac{W(d,t)}{|\Xi(t-\mathrm{i}d)|^2}=\frac{R(d,t)I_t(d,t)-I(d,t)R_t(d,t)}{R(d,t)^2+I(d,t)^2}=\frac{\mathrm{Re}(\overline{\xi(s)}\,\xi'(s))}{|\xi(s)|^2}=\mathrm{Re}\left(\frac{\xi'(\frac{1}{2}+d+\mathrm{i}t)}{\xi(\frac{1}{2}+d+\mathrm{i}t)}\right). \tag{1.7}$$

The nontrivial zeros of $\zeta(s)$ are written as $\rho=\beta_\rho+\mathrm{i}\gamma_\rho=\frac{1}{2}+d_\rho+\mathrm{i}\gamma_\rho$, where $d_\rho=\beta_\rho-\frac{1}{2}\in(-\frac{1}{2},\frac{1}{2})$ is the off-critical displacement ($d_\rho=0$ on the critical line). The first three positive critical-line ordinates are $\gamma_1\approx 14.13472514$, $\gamma_2\approx 21.02203964$, and $\gamma_3\approx 25.01085758$. We denote the digamma function by $\psi(w):=\Gamma'(w)/\Gamma(w)$ and the von Mangoldt function by $\Lambda(n)$.

2. Pure-Square Trigonometric Wronskian Multiplets and Fejér–Lukács Pole Suppression

We begin by deriving the exact logarithmic-derivative decomposition of $\widehat{W}(d,t)$ and establishing the Fejér–Lukács Toeplitz optimization theorem for pure-square Wronskian multiplets on $d>\frac{1}{2}$.

Lemma 2.1 (Logarithmic Derivative Decomposition of $\widehat{W}(d,t)$)**.** *For all $d>0$ and $t\in\mathbb{R}$ at which $\zeta(\frac{1}{2}+d+\mathrm{i}t)\neq 0$, the normalized transverse Wronskian (1.7) decomposes as*

$$\widehat{W}(d,t)=\mathcal{A}(d,t)-\mathcal{Z}(d,t),\qquad \mathcal{Z}(d,t):=-\mathrm{Re}\left(\frac{\zeta'(\frac{1}{2}+d+\mathrm{i}t)}{\zeta(\frac{1}{2}+d+\mathrm{i}t)}\right), \tag{2.1}$$

where the total Archimedian carrier $\mathcal{A}(d,t):=\mathcal{A}_\Gamma(d,t)+\mathcal{P}_{\mathrm{pole}}(d,t)$ consists of the digamma drift and polar terms

$$\mathcal{A}_\Gamma(d,t):=\frac{1}{2}\mathrm{Re}\,\psi\left(\frac{1}{4}+\frac{d}{2}+\mathrm{i}\frac{t}{2}\right)-\frac{1}{2}\ln\pi,\qquad \mathcal{P}_{\mathrm{pole}}(d,t):=\frac{d+\frac{1}{2}}{(d+\frac{1}{2})^2+t^2}+\frac{d-\frac{1}{2}}{(d-\frac{1}{2})^2+t^2}. \tag{2.2}$$

For $d>\frac{1}{2}$ ($\sigma>1$), the Dirichlet series $\mathcal{Z}(d,t)=\sum_{n=2}^\infty\Lambda(n)n^{-(1/2+d)}\cos(t\ln n)$ converges absolutely and is bounded in modulus by its zero-height majorant

$$|\mathcal{Z}(d,t)|\leq M(d):=-\frac{\zeta'(\frac{1}{2}+d)}{\zeta(\frac{1}{2}+d)}=\sum_{n=2}^\infty\frac{\Lambda(n)}{n^{1/2+d}}=\frac{1}{d-\frac{1}{2}}-\gamma_{\mathrm{E}}+\mathcal{O}\left(d-\frac{1}{2}\right). \tag{2.3}$$

Proof. Differentiating $\ln\xi(s)=-\ln 2+\ln s+\ln(s-1)-\frac{s}{2}\ln\pi+\ln\Gamma(s/2)+\ln\zeta(s)$ at $s=\frac{1}{2}+d+\mathrm{i}t$ away from the zeros and pole of $\zeta(s)$ and taking the real part via (1.7) yields (2.1)–(2.2). For $d>\frac{1}{2}$, the Euler product $\zeta(s)=\prod_p(1-p^{-s})^{-1}$ converges absolutely, so termwise logarithmic

differentiation gives $-\zeta'(s)/\zeta(s) = \sum_{n=2}^{\infty} \Lambda(n) n^{-s}$, whose real part is $\mathcal{Z}(d,t)$ and whose modulus is bounded by the positive series $M(d) = \mathcal{Z}(d,0)$ via $|\cos(t \ln n)| \le 1$. The Laurent expansion of $-\zeta'(\sigma)/\zeta(\sigma)$ at $\sigma = 1$ gives the final estimate in (2.3). □

To eliminate the sign oscillations of $\cos(kt \ln n)$ across all prime powers simultaneously, we couple the harmonics $\widehat{W}(d, kt)$ $(1 \le k \le K)$ with a non-negative trigonometric polynomial.

Theorem 2.2 (Pure-Square Trigonometric Multiplets and Fejér–Lukács Pole Suppression)**.** *Let $K \ge 1$ and let $(a_0, a_1, \dots, a_K) \in \mathbb{R}^{K+1} \setminus \{0\}$ with $a_j \ge 0$. Define the non-negative trigonometric polynomial*

$$T_K(\theta) := \left| \sum_{j=0}^{K} a_j \, \mathrm{e}^{\mathrm{i}j\theta} \right|^2 = c_0 + \sum_{k=1}^{K} c_k \cos(k\theta) \ge 0, \qquad c_0 := \sum_{j=0}^{K} a_j^2 > 0, \quad c_k := 2 \sum_{j=0}^{K-k} a_j a_{j+k} \ge 0. \tag{2.4}$$

Then for all $d > \frac{1}{2}$ and $t \in \mathbb{R}$:

(i) ***Dual Pure-Square Identities.*** *The pure-square prime sums*

$$\boxed{\mathcal{S}_{\pm}(d,t) := c_0 M(d) \pm \sum_{k=1}^{K} (\pm 1)^{k-1} c_k \mathcal{Z}(d,kt) \equiv \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^{1/2+d}} \left| \sum_{j=0}^{K} (\pm 1)^j a_j \, n^{\mathrm{i}jt} \right|^2 \ge 0} \tag{2.5}$$

satisfy the exact dual identities (1.1). In particular, $\sum_{k=1}^{K} c_k \widehat{W}(d,kt) \le \sum_{k=1}^{K} c_k \mathcal{A}(d,kt) + c_0 M(d)$ and $\sum_{k=1}^{K} (-1)^{k-1} c_k \widehat{W}(d,kt) \ge \sum_{k=1}^{K} (-1)^{k-1} c_k \mathcal{A}(d,kt) - c_0 M(d)$ for all $d > \frac{1}{2}$ and $t \in \mathbb{R}$.

(ii) ***Fejér–Lukács Sine-Eigenvector Theorem and $\sigma = 1$ Pole Suppression.*** *Over all amplitude vectors $\mathbf{a} = (a_0, \dots, a_K) \in \mathbb{R}^{K+1} \setminus \{0\}$, the fundamental first-harmonic ratio c_1/c_0 satisfies the sharp Rayleigh bound*

$$\max_{\mathbf{a} \ne 0} \frac{c_1(\mathbf{a})}{c_0(\mathbf{a})} = \Lambda_{\max}(K) := 2 \cos\!\left(\frac{\pi}{K+2} \right) \in [1, 2), \tag{2.6}$$

attained uniquely (up to scaling) by the Fejér–Lukács sine eigenvector $a_j^{\mathrm{FL}} := \sin(\frac{(j+1)\pi}{K+2})$ $(0 \le j \le K)$. For this eigenvector, $c_0^{\mathrm{FL}} = \frac{K+2}{2}$, $c_1^{\mathrm{FL}} = (K+2)\cos(\frac{\pi}{K+2})$, and the total Archimedian-to-pole leverage ratio is given in closed form by

$$\frac{1}{c_0^{\mathrm{FL}}} \sum_{k=1}^{K} c_k^{\mathrm{FL}} = \frac{2}{K+2} \cot^2\!\left(\frac{\pi}{2(K+2)} \right) - 1 = \frac{8}{\pi^2} (K+2) - 1 + \mathcal{O}\!\left(\frac{1}{K} \right). \tag{2.7}$$

Proof. (i) For $d > \frac{1}{2}$, the Dirichlet series for $M(d)$ and $\mathcal{Z}(d,kt)$ converge absolutely by Lemma 2.1. By Fubini's theorem for finite linear combinations of absolutely convergent series, we may interchange the finite sum over $k \in \{1, \dots, K\}$ with the infinite sum over $n \ge 2$:

$$c_0 M(d) + \sum_{k=1}^{K} c_k \mathcal{Z}(d,kt) = \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^{1/2+d}} \left[c_0 + \sum_{k=1}^{K} c_k \cos(kt \ln n) \right] = \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^{1/2+d}} \, T_K(t \ln n) \ge 0,$$

where the final equality uses $\left| \sum_{j=0}^{K} a_j \mathrm{e}^{\mathrm{i}j\theta} \right|^2 = \sum_{j=0}^{K} a_j^2 + 2 \sum_{k=1}^{K} \sum_{j=0}^{K-k} a_j a_{j+k} \cos(k\theta) = c_0 + \sum_{k=1}^{K} c_k \cos(k\theta)$. Substituting $\mathcal{Z}(d,kt) = \mathcal{A}(d,kt) - \widehat{W}(d,kt)$ from (2.1) yields the first identity of (1.1). Evaluating the same relation at the shifted phase $\theta \mapsto \theta + \pi$, for which $\cos(k(\theta+\pi)) = (-1)^k \cos(k\theta)$ and $\mathrm{e}^{\mathrm{i}j(\theta+\pi)} = (-1)^j \mathrm{e}^{\mathrm{i}j\theta}$, yields the second identity of (1.1) and (2.5).

(ii) By definition (2.4), $c_1(\mathbf{a})/c_0(\mathbf{a}) = (\mathbf{a}^{\mathsf{T}} \mathbf{J}_K \mathbf{a})/(\mathbf{a}^{\mathsf{T}} \mathbf{a})$, where $\mathbf{J}_K$ is the $(K+1) \times (K+1)$ symmetric tridiagonal Toeplitz adjacency matrix with $(\mathbf{J}_K)_{j,m} = 1$ if $|j-m| = 1$ and 0 otherwise. As shown in Appendix A, the eigenvalues of $\mathbf{J}_K$ are $\lambda_r = 2\cos(\frac{r\pi}{K+2})$ for $r = 1, \dots, K+1$, with corresponding

TABLE 1. 50-decimal `mpmath` and 35-digit `mpmath.iv` certification of Theorem 2.2 at $d = 0.65$ ($\sigma = 1.15 > 1$) and $t = \gamma_3 = 25.01085758\ldots$, where $M(0.65) = 6.11646754355$.

Family	K	c_0	c_1	Ratio c_1/c_0	Pole Ratio $(\sum c_k)/c_0$	$\mathcal{S}_+(0.65,\gamma_3) \geq 0$	50-dps Err
`dVP_K2`	2	3.000000	4.000000	1.33333333	1.66666667	13.93490986	1.93×10^{-51}
`Opt_K2`	2	4.199615	5.932438	1.41261472	1.88884639	18.82520514	0.00×10^{0}
`Opt_K4`	4	22.38889	38.17518	1.70509493	3.26340282	88.92975741	1.69×10^{-51}
`Fejer_K6`	6	4.000000	7.391036	1.84775907	5.31853559	15.94902438	0.00×10^{0}

orthogonal eigenvectors $v_j^{(r)} = \sin(\frac{r(j+1)\pi}{K+2})$. The largest eigenvalue is $\lambda_1 = 2\cos(\frac{\pi}{K+2})$ for $r = 1$, which establishes (2.6). Moreover, evaluating $T_K(0) = c_0^{\mathrm{FL}} + \sum_{k=1}^{K} c_k^{\mathrm{FL}} = \big(\sum_{j=0}^{K} \sin\frac{(j+1)\pi}{K+2}\big)^2 = \cot^2(\frac{\pi}{2(K+2)})$ and dividing by $c_0^{\mathrm{FL}} = \frac{K+2}{2}$ (proved in Appendix A) establishes (2.7). □

Table 1 records the 50-decimal `mpmath` and 35-digit `mpmath.iv` certification of Theorem 2.2 at $(d,t) = (0.65,\gamma_3)$, where $M(0.65) = 6.11646754355\ldots$, across the classical de la Vallée Poussin polynomial (`dVP_K2`), the Kondratiev-optimized degree-2 and degree-4 polynomials (`Opt_K2`, `Opt_K4`), and the degree-6 Fejér–Lukács polynomial (`Fejer_K6`).

3. The Mirror-Paired Hadamard Square-Denominator Identity and Dual Sieved Pure-Square Decomposition

We now connect the trigonometric identities (1.1) to the spectral side by pairing each nontrivial zero ρ with its functional-equation reflection $1-\bar\rho$ across the critical line.

Theorem 3.1 (Mirror-Paired Hadamard Square-Denominator Identity)**.** *For every nontrivial zero $\rho = \frac12 + d_\rho + \mathrm{i}\gamma_\rho$ of $\zeta(s)$ with $d_\rho \in (-\frac12,\frac12)$, its reflection across the critical line at the same ordinate γ_ρ is $1-\bar\rho = \frac12 - d_\rho + \mathrm{i}\gamma_\rho$. Setting $u := t - \gamma_\rho \in \mathbb{R}$, the mean transverse Hadamard kernel (1.2) of the pair $\{\rho, 1-\bar\rho\}$ satisfies the exact algebraic identities*

$$K(d,d_\rho,u) = \frac{d\,(d^2 - d_\rho^2 + u^2)}{D(d,d_\rho,u)}, \tag{3.1}$$

where the denominator $D(d,d_\rho,u) := ((d-d_\rho)^2+u^2)((d+d_\rho)^2+u^2)$ admits the two equivalent pure-square decompositions

$$D(d,d_\rho,u) \equiv (d^2 - d_\rho^2 + u^2)^2 + 4u^2 d_\rho^2 \equiv (d^2 + d_\rho^2 + u^2)^2 - 4d^2 d_\rho^2. \tag{3.2}$$

Consequently:

(i) *For all $d > 0$, $d_\rho \in (-\frac12,\frac12)$, and $u \neq 0$, we have $D(d,d_\rho,u) > 0$.*

(ii) *Whenever $d > |d_\rho|$ (in particular, throughout the Euler-product half-plane $d \geq \frac12 > |d_\rho|$), the kernel satisfies $K(d,d_\rho,u) > 0$ for all $u \in \mathbb{R}$, with central value $K(d,d_\rho,0) = \frac{d}{d^2-d_\rho^2} \geq \frac{1}{d}$.*

(iii) *Inside the critical strip $0 < d \leq \frac12$, we still have $K(d,d_\rho,u) > 0$ at every vertical separation satisfying $|u| > \sqrt{\max(0, d_\rho^2 - d^2)}$ (in particular, for all $|u| \geq \frac12$ independently of d_ρ).*

Proof. Putting the two fractions in (1.2) over the common denominator $D(d,d_\rho,u) = ((d-d_\rho)^2 + u^2)((d+d_\rho)^2+u^2)$ gives the numerator

$$\frac12\Big[(d-d_\rho)((d+d_\rho)^2+u^2) + (d+d_\rho)((d-d_\rho)^2+u^2)\Big] = \frac12\Big[2d(d^2-d_\rho^2) + 2du^2\Big] = d(d^2 - d_\rho^2 + u^2),$$

which proves (3.1). Writing $(d \mp d_\rho)^2 + u^2 = (d^2 + d_\rho^2 + u^2) \mp 2dd_\rho$ and multiplying the conjugate factors yields the second equivalence in (3.2):

$$D(d,d_\rho,u) = (d^2+d_\rho^2+u^2)^2 - 4d^2d_\rho^2 = (d^2-d_\rho^2)^2 + 2u^2(d^2+d_\rho^2) + u^4 = (d^2 - d_\rho^2 + u^2)^2 + 4u^2d_\rho^2,$$

which proves the first equivalence in (3.2). Since $D(d, d_\rho, u)$ is a sum of two squares in which $4u^2 d_\rho^2 > 0$ if $u \neq 0$ and $d_\rho \neq 0$, while $D(d, 0, u) = (d^2 + u^2)^2 > 0$, part (i) holds. Parts (ii) and (iii) follow immediately from the sign of the numerator factor $d^2 - d_\rho^2 + u^2$. □

Remark 3.2 (Connection with the Guinand–Weil explicit formula and Sylvester signature). In the explicit formula of Guinand [6] and Weil [28], a test function $f(u)$ and its Fourier–Mellin transform $\widehat{f}(s)$ pair the prime powers $\Lambda(n)n^{-1/2}$ against the spectral sum $\sum_\rho \widehat{f}(\rho)$. When tested against the Poisson–Cauchy family parametrized by transverse distance $d > 0$, an on-critical zero ($d_\rho = 0$) generates the single Lorentzian $d/(d^2 + u^2)$, whereas an off-critical mirror pair $\{\frac{1}{2} + d_\rho + \mathrm{i}\gamma_\rho, \frac{1}{2} - d_\rho + \mathrm{i}\gamma_\rho\}$ with $d_\rho \neq 0$ splits the Lorentzian pole into two shifted poles at $d = \pm d_\rho$. By (3.1), for $0 < d < |d_\rho|$ the numerator $d(d^2 - d_\rho^2 + u^2)$ becomes strictly negative on the focal interval $|u| < \sqrt{d_\rho^2 - d^2}$, creating a negative Sylvester signature defect at $u = 0$ ($K(d, d_\rho, 0) = -\frac{d}{d_\rho^2 - d^2} < 0$), whereas for $d > |d_\rho|$ both the numerator and the square denominator $D(d, d_\rho, u)$ are strictly positive for all $u \in \mathbb{R}$.

We now combine Theorem 2.2(i) and Theorem 3.1 with a finite prime-power sieve at cutoff $X \geq 1$ to obtain an exact identity and monotone critical-strip boundary $d^{\mathrm{mirror}}_{\mathrm{strip},X}(t)$.

Theorem 3.3 (Dual Sieved Pure-Square Decomposition and Monotone Critical-Strip Penetration). *Let $T_K(\theta) = \left|\sum_{j=0}^K a_j \mathrm{e}^{\mathrm{i}j\theta}\right|^2 = c_0 + \sum_{k=1}^K c_k \cos(k\theta) \geq 0$ with $c_1 > c_0 > 0$ and $c_k \geq 0$. For any cutoff $X \geq 1$ and $d > \frac{1}{2}$, split $\mathcal{S}_+(d, t) = \Delta B_X(d, t) + \mathcal{S}_{+,>X}(d, t)$ into the finite pure-square prime sum and non-negative tail*

$$\Delta B_X(d,t) := \sum_{n=2}^{\lfloor X \rfloor} \frac{\Lambda(n)}{n^{1/2+d}} \left| \sum_{j=0}^K a_j\, n^{\mathrm{i}jt} \right|^2 \geq 0, \qquad \mathcal{S}_{+,>X}(d,t) := \sum_{n>X} \frac{\Lambda(n)}{n^{1/2+d}} \left| \sum_{j=0}^K a_j\, n^{\mathrm{i}jt} \right|^2 \geq 0. \tag{3.3}$$

Then:

(i) ***Dual Sieved Pure-Square Identity.*** *For all $d > \frac{1}{2}$, $t \in \mathbb{R}$, and $X \geq 1$, we have the exact identity*

$$\sum_{k=1}^K c_k\, \widehat{W}(d, kt) + \mathcal{S}_{+,>X}(d,t) \equiv U_{T,X}(d,t) := \sum_{k=1}^K c_k\, \mathcal{A}(d, kt) + c_0 M(d) - \Delta B_X(d,t). \tag{3.4}$$

(ii) ***Monotone Critical-Strip Penetration.*** *Suppose $\rho_0 = \frac{1}{2} + d_{\rho_0} + \mathrm{i}t$ is a nontrivial zero of $\zeta(s)$ at ordinate $t > 0$ with off-critical displacement $d_{\rho_0} \in (-\frac{1}{2}, \frac{1}{2})$. Then for every $X \geq 1$, $|d_{\rho_0}|$ is bounded above by the sieved mirror-paired threshold*

$$|d_{\rho_0}| \leq d^{\mathrm{mirror}}_{\mathrm{strip},X}(t) := \inf_{d > 1/2} \sqrt{\max\left(0,\, d^2 - \frac{2c_1 d}{U_{T,X}(d,t)}\right)}. \tag{3.5}$$

Moreover, $d^{\mathrm{mirror}}_{\mathrm{strip},X}(t)$ is monotonically non-increasing in X: for all $1 \leq X_1 \leq X_2$,

$$0 \leq d^{\mathrm{mirror}}_{\mathrm{strip},X_2}(t) \leq d^{\mathrm{mirror}}_{\mathrm{strip},X_1}(t) \leq d^{\mathrm{mirror}}_{\mathrm{strip},1}(t) < \frac{1}{2}. \tag{3.6}$$

Proof. (i) Substituting $\mathcal{S}_+(d,t) = \Delta B_X(d,t) + \mathcal{S}_{+,>X}(d,t)$ into the first identity of (1.1) and subtracting $\Delta B_X(d,t)$ from both sides yields (3.4).

(ii) By Hadamard's partial-fraction expansion and Theorem 3.1(ii), for every $d > \frac{1}{2} > |d_\rho|$ we may group the nontrivial zeros into mirror pairs $\{\rho, 1 - \bar{\rho}\}$ (where a critical-line zero with $d_\rho = 0$ equals its own mirror and contributes $K(d, 0, u) = \frac{d}{d^2 + u^2} > 0$, while an off-critical zero with $d_{\rho_0} \neq 0$ has a distinct mirror $1 - \bar{\rho}_0 \neq \rho_0$ at the same ordinate t and together they contribute $2K(d, d_{\rho_0}, 0) = \frac{2d}{d^2 - d_{\rho_0}^2}$

TABLE 2. Monotone critical-strip penetration $d^{\mathrm{mirror}}_{\mathrm{strip},X}(t)$ from Theorem 3.3 for the degree-4 multiplet `Opt_K4` across $X \in \{1, 10, 50, 200, 1000\}$, together with the naive Euler wall $d_{\mathrm{Euler}}(t)$ ($\mathcal{A}(d,t) = M(d)$) and the multi-multiplet envelope over $K \in \{2, 4, 6\}$ at $X = 1000$.

Height t	$d_{\mathrm{Euler}}(t)$	$X = 1$ (`Opt_K4`)	$X = 10$	$X = 50$	$X = 200$	$X = 1000$	Envelope $K \in \{2,4,6\}$
$\gamma_1 \approx 14.13$	1.703294	0.163626	0.000000	0.000000	0.000000	0.000000	0.000000
$\gamma_2 \approx 21.02$	1.456763	0.372173	0.345634	0.136242	0.000000	0.000000	0.000000
$\gamma_3 \approx 25.01$	1.377825	0.402016	0.399237	0.369932	0.325316	0.289068	0.000000
40.00	1.213058	0.440891	0.427051	0.378283	0.322923	0.220671	0.220671
60.00	1.122793	0.457962	0.447783	0.386527	0.334392	0.283309	0.000000
100.00	1.034139	0.469018	0.360817	0.223455	0.122324	0.000000	0.000000
250.00	0.930921	0.478926	0.465047	0.434292	0.391093	0.324943	0.264026
1000.00	0.827684	0.486061	0.485282	0.477908	0.472748	0.465837	0.409656
10000.00	0.728102	0.490650	0.490400	0.489217	0.487879	0.486737	0.483231

to $\widehat{W}(d,t)$). Since $K(d, d_\rho, kt - \gamma_\rho) > 0$ for all zeros at all harmonics $1 \le k \le K$, and since $c_k \ge 0$ and $\mathcal{S}_{+,>X}(d,t) \ge 0$, identity (3.4) implies for any off-critical zero pair at height t:

$$\frac{2c_1 d}{d^2 - d_{\rho_0}^2} \le c_1 \widehat{W}(d,t) \le \sum_{k=1}^{K} c_k\, \widehat{W}(d,kt) + \mathcal{S}_{+,>X}(d,t) = U_{T,X}(d,t) \qquad \left(\forall\, d > \frac{1}{2}\right).$$

Solving for $d_{\rho_0}^2$ gives $d_{\rho_0}^2 \le d^2 - \frac{2c_1 d}{U_{T,X}(d,t)}$ for every $d > \frac{1}{2}$, which proves (3.5). Since every term in $\Delta B_X(d,t)$ is non-negative, $X_2 \ge X_1 \implies \Delta B_{X_2}(d,t) \ge \Delta B_{X_1}(d,t) \implies 0 < U_{T,X_2}(d,t) \le U_{T,X_1}(d,t)$, which implies $d^{\mathrm{mirror}}_{\mathrm{strip},X_2}(t) \le d^{\mathrm{mirror}}_{\mathrm{strip},X_1}(t)$. Finally, as $d \to (\frac{1}{2})^+$, $M(d) = (d - \frac{1}{2})^{-1} + \mathcal{O}(1)$ gives $U_{T,1}(d,t) = c_0(d - \frac{1}{2})^{-1} + \mathcal{O}(1)$, so at $d = \frac{1}{2} + \varepsilon$ with $\varepsilon > 0$ small we have

$$d^2 - \frac{2c_1 d}{U_{T,1}(d,t)} = \frac{1}{4} + \left(1 - \frac{c_1}{c_0}\right)\varepsilon + \mathcal{O}(\varepsilon^2) < \frac{1}{4}$$

because $c_1 > c_0$. Taking the infimum over $d > \frac{1}{2}$ yields $d^{\mathrm{mirror}}_{\mathrm{strip},1}(t) < \frac{1}{2}$, completing the proof of (3.6). □

4. The Universal Conservation Law $\int_{\mathbb{R}} K\,\mathrm{d}u \equiv \pi$ and the Hypothesis-Free Minorant $K_{\min}(d,u)$

In Theorem 3.3(ii), every background zero pair at ordinate $\gamma_\rho \ne kt$ was bounded below by zero ($K(d, d_\rho, kt - \gamma_\rho) > 0$). In reality, the dense sea of nontrivial zeros of $\zeta(s)$ has asymptotic density $\frac{1}{2\pi}\ln(\gamma/2\pi)$ and carries a strictly positive collective weight in every harmonic channel $\widehat{W}(d,kt)$. To quantify this background contribution without circularly assuming anything about the unknown abscissas $d_\rho \in (-\frac{1}{2}, \frac{1}{2})$, we establish two structural properties of $K(d, d_\rho, u)$: its conserved π-integral and its exact difference factorization relative to the strip boundary $d_\rho \to \frac{1}{2}$.

Theorem 4.1 (Universal Conservation Law $\int_{\mathbb{R}} K\,\mathrm{d}u \equiv \pi$ and Exact Window Antiderivative)**.** *For every transverse distance $d > 0$ and every zero displacement $d_\rho \in (-\frac{1}{2}, \frac{1}{2})$ satisfying $d > |d_\rho|$, the mirror-paired Hadamard kernel (1.2) is strictly positive and integrable on $\mathbb{R}$. For any window half-width $H \ge 0$, its symmetric integral over $[-H, H]$ is given in closed form by*

$$\int_{-H}^{H} K(d, d_\rho, u)\,\mathrm{d}u = \arctan\left(\frac{H}{d - d_\rho}\right) + \arctan\left(\frac{H}{d + d_\rho}\right), \tag{4.1}$$

Paper VII — Pure-Square Trigonometric Multiplets, Fejér–Lukács Pole Suppression & Monotone Sieved Strip Penetration

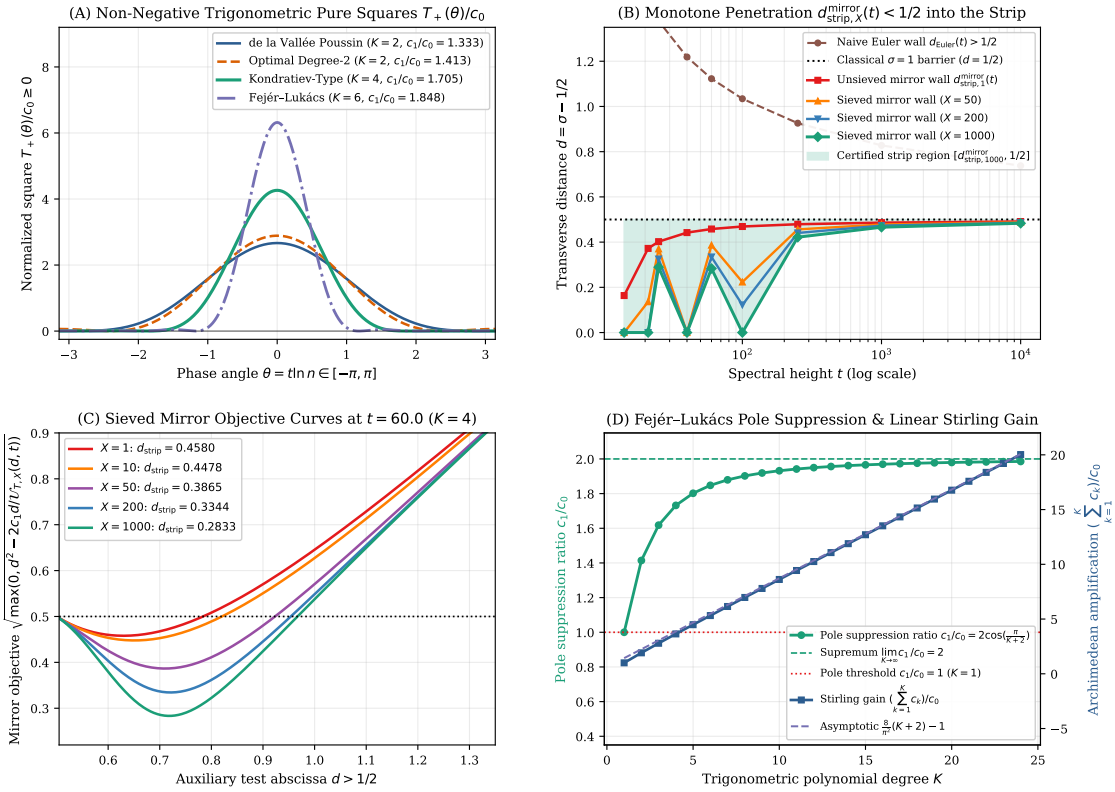


FIGURE 1. **(A)** Non-negative trigonometric pure squares $T_K(\theta)/c_0$ for `dVP_K2`, `Opt_K2`, `Opt_K4`, and `Fejer_K6`. **(B)** Monotone critical-strip penetration $d^{\text{mirror}}_{\text{strip},X}(t) < \frac{1}{2}$ across $X \in \{1, 50, 200, 1000\}$ versus the naive Euler wall $d_{\text{Euler}}(t) > \frac{1}{2}$. **(C)** Sieved mirror objective curves $\sqrt{\max(0, d^2 - 2c_1 d/U_{T,X}(d,t))}$ at $t = 60.0$ ($K = 4$). **(D)** Fejér–Lukács Toeplitz spectral maximum $\Lambda_{\max}(K) = 2\cos(\frac{\pi}{K+2}) \to 2$ and linear Archimedean amplification $(\sum_{k=1}^{K} c_k^{\text{FL}})/c_0^{\text{FL}} = \frac{2}{K+2}\cot^2(\frac{\pi}{2(K+2)}) - 1$.

and its improper integral over the entire vertical line $\mathbb{R}$ *satisfies the universal conservation law*

$$I(d, d_\rho) := \int_{-\infty}^{\infty} K(d, d_\rho, u)\,\mathrm{d}u \equiv \pi \qquad (\forall\, d > |d_\rho|,\ d_\rho \in (-1/2, 1/2)), \tag{4.2}$$

independently of both the interrogation distance d *and the unknown zero abscissa* d_ρ.

Proof. Since $d > |d_\rho|$, both shifted distances $a_- := d - d_\rho > 0$ and $a_+ := d + d_\rho > 0$ are strictly positive. Consequently, each Cauchy–Lorentz summand $\frac{a_\pm}{a_\pm^2 + u^2}$ in the partial-fraction definition (1.2) of $K(d, d_\rho, u)$ is smooth, even, strictly positive on $\mathbb{R}$, and decays as $a_\pm u^{-2}$ as $|u| \to \infty$, ensuring absolute integrability on $\mathbb{R}$. Integrating term by term on $[-H, H]$ via the elementary antiderivative $\int \frac{a}{a^2+u^2}\,\mathrm{d}u = \arctan(u/a)$ for $a > 0$ gives

$$\int_{-H}^{H} K(d, d_\rho, u)\,\mathrm{d}u = \frac{1}{2}\left[\arctan\left(\frac{u}{a_-}\right) + \arctan\left(\frac{u}{a_+}\right)\right]_{-H}^{H} = \arctan\left(\frac{H}{d - d_\rho}\right) + \arctan\left(\frac{H}{d + d_\rho}\right),$$

which proves (4.1). Taking the limit $H \to \infty$ and using $\lim_{x\to+\infty} \arctan(x) = \frac{\pi}{2}$ since $d - d_\rho > 0$ and $d + d_\rho > 0$, we obtain $I(d, d_\rho) = \frac{\pi}{2} + \frac{\pi}{2} \equiv \pi$, establishing (4.2). □

While the total integral $\int_{\mathbb{R}} K(d, d_\rho, u)\,\mathrm{d}u \equiv \pi$ is independent of d_ρ, the pointwise profile $K(d, d_\rho, u)$ depends on d_ρ^2. The following theorem proves that in the far field $|u| \geq H_{\min}(d)$, $K(d, d_\rho, u)$ is monotonically decreasing in $d_\rho^2 \in [0, \frac{1}{4})$, so the strip-boundary limit $d_\rho \to \frac{1}{2}$ yields an unconditional pointwise minorant.

Theorem 4.2 (Exact Difference Factorization and Hypothesis-Free Minorant $K_{\min}(d,u)$)**.** *For all $d > 0$ and $u \in \mathbb{R} \setminus \{0\}$, define the strip-boundary minorant kernel*

$$K_{\min}(d,u) := \lim_{|d_\rho| \to 1/2} K(d, d_\rho, u) = \frac{d\,(d^2 - \frac{1}{4} + u^2)}{(d^2 - \frac{1}{4} + u^2)^2 + u^2} = \frac{1}{2}\left[\frac{d - \frac{1}{2}}{(d - \frac{1}{2})^2 + u^2} + \frac{d + \frac{1}{2}}{(d + \frac{1}{2})^2 + u^2}\right]. \tag{4.3}$$

Then:

(i) ***Exact Difference Factorization and Partial Derivative.*** *Putting $K(d, d_\rho, u) - K_{\min}(d, u)$ over its common denominator with $W := d^2 + u^2$ yields the exact algebraic factorization*

$$\begin{aligned} K(d, d_\rho, u) - K_{\min}(d, u) &\equiv \frac{d\,(\frac{1}{4} - d_\rho^2)\,[4u^2 W - (W - d_\rho^2)(W - \frac{1}{4})]}{((W - d_\rho^2)^2 + 4u^2 d_\rho^2)\,((W - \frac{1}{4})^2 + u^2)} \\ &\equiv \frac{d\,(\frac{1}{4} - d_\rho^2)\,[3u^4 + (2d^2 + d_\rho^2 + \frac{1}{4})u^2 - (d^2 - \frac{1}{4})(d^2 - d_\rho^2)]}{((d^2 - d_\rho^2 + u^2)^2 + 4u^2 d_\rho^2)\,((d^2 - \frac{1}{4} + u^2)^2 + u^2)}, \end{aligned} \tag{4.4}$$

while differentiating $K(d, d_\rho, u)$ with respect to $v := d_\rho^2 \in [0, \frac{1}{4})$ gives

$$-\frac{\partial K}{\partial (d_\rho^2)}(d, d_\rho, u) \equiv \frac{d\,[4u^2 W - (W - d_\rho^2)^2]}{((W - d_\rho^2)^2 + 4u^2 d_\rho^2)^2} = \frac{d\,[3u^4 + 2(d^2 + d_\rho^2)u^2 - (d^2 - d_\rho^2)^2]}{((d^2 - d_\rho^2 + u^2)^2 + 4u^2 d_\rho^2)^2}. \tag{4.5}$$

(ii) ***Universal Far-Field Minorant Property.*** *For all $d \geq \frac{1}{2}$ (the Euler-product half-plane where the minorant is applied in Section 5), all $d_\rho \in (-\frac{1}{2}, \frac{1}{2})$, and all vertical separations $|u|$ ($u \neq 0$) satisfying*

$$|u| \geq H_{\min}(d) := \sqrt{\frac{d^2 - 1/4}{3}}, \tag{4.6}$$

we have the hypothesis-free secant lower bound

$$K(d, d_\rho, u) \geq K_{\min}(d, u) > 0 \qquad (\forall\, d \geq \tfrac{1}{2},\ d_\rho \in (-1/2, 1/2),\ |u| \geq H_{\min}(d),\ u \neq 0). \tag{4.7}$$

Inside the critical strip $0 < d < \frac{1}{2}$, the secant inequality $K(d, d_\rho, u) \geq K_{\min}(d, u) > 0$ holds for all $|u| > \sqrt{\frac{1}{4} - d^2}$ (where $W - \frac{1}{4} > 0$), whereas the pointwise derivative inequality $-\partial K/\partial(d_\rho^2) \geq 0$ in (4.5) holds at $d_\rho = 0$ for all $|u| \geq d/\sqrt{3}$.

(iii) ***Monotone Vertical Decay.*** *For every $d > \frac{1}{2}$, $K_{\min}(d, u)$ is strictly decreasing in $|u|$ on $|u| \geq H_{\min}(d)$, with explicit derivative for $u > 0$:*

$$\frac{\partial K_{\min}}{\partial u}(d, u) = -u\left[\frac{d - \frac{1}{2}}{((d - \frac{1}{2})^2 + u^2)^2} + \frac{d + \frac{1}{2}}{((d + \frac{1}{2})^2 + u^2)^2}\right] < 0. \tag{4.8}$$

Proof. (i) Let $W := d^2 + u^2$ and $v := d_\rho^2 \in [0, \frac{1}{4})$. By (3.1) and (3.2), $K(d, d_\rho, u) = \frac{d(W - v)}{(W - v)^2 + 4u^2 v}$ and $K_{\min}(d, u) = \frac{d(W - 1/4)}{(W - 1/4)^2 + u^2}$. Forming their difference over the positive common denominator $((W - v)^2 + 4u^2 v)((W - \frac{1}{4})^2 + u^2)$ yields the numerator (Detailed algebraic verification is given in Appendix B):

$$\begin{aligned} \mathcal{N} &:= d\Big[(W - v)((W - \tfrac{1}{4})^2 + u^2) - (W - \tfrac{1}{4})((W - v)^2 + 4u^2 v)\Big] \\ &= d\Big[(W - v)(W - \tfrac{1}{4})((W - \tfrac{1}{4}) - (W - v)) + u^2((W - v) - 4v(W - \tfrac{1}{4}))\Big] \end{aligned}$$

TABLE 3. 50-decimal `mpmath` and 35-digit `mpmath.iv` certification of the difference factorization (4.4) and the universal conservation law $\int_{\mathbb{R}} K(d, d_\rho, u)\,\mathrm{d}u \equiv \pi$ across benchmark triples.

Theorem 4.2: Difference Factorization $K - K_{\min} \geq 0$						Theorem 4.1: π-Conservation and Tail Mass				
(d, d_ρ, u)	$K(d, d_\rho, u)$	$K_{\min}(d, u)$	$K - K_{\min}$	Fact. Err	`iv`	(d, d_ρ, H)	$\lvert\int_{\mathbb{R}} K - \pi\rvert$	$\pi\alpha_{\text{tail}}$	`iv` Width	$\pi \in I_{\text{iv}}$
$(0.55, 0.00, 0.25)$	1.505818	1.045455	0.460363	0.00	PASS	$(0.55, 0.00, 0.50)$	0.00	0.916931	2.20e-18	PASS
$(0.55, 0.35, 0.50)$	1.236742	1.007979	0.228764	3.51e-51	PASS	$(0.55, 0.48, 1.00)$	0.00	0.517739	2.06e-18	PASS
$(0.65, 0.20, 0.30)$	1.318389	0.768981	0.549408	0.00	PASS	$(0.65, 0.25, 1.00)$	0.00	0.840035	2.60e-18	PASS
$(0.65, 0.45, 1.00)$	0.520521	0.508652	0.011869	4.16e-51	PASS	$(0.75, 0.40, 2.00)$	0.00	0.592408	3.00e-18	PASS
$(0.75, 0.10, 0.50)$	0.918200	0.800000	0.118200	4.23e-51	PASS	$(1.00, 0.49, 3.00)$	0.00	0.596410	4.00e-18	PASS
$(1.10, 0.40, 1.50)$	0.320462	0.320182	0.000280	3.38e-52	PASS	$(1.35, 0.15, 4.00)$	0.00	0.630519	5.40e-18	PASS

$$= d\Big[(v - \tfrac{1}{4})(W - v)(W - \tfrac{1}{4}) + u^2(1 - 4v)W\Big]$$
$$= d\,(\tfrac{1}{4} - v)\big[4u^2W - (W - v)(W - \tfrac{1}{4})\big].$$

Substituting $W = d^2 + u^2$ into the bracket gives

$$4u^2(d^2 + u^2) - (d^2 - v + u^2)(d^2 - \tfrac{1}{4} + u^2) = 3u^4 + (2d^2 + v + \tfrac{1}{4})u^2 - (d^2 - \tfrac{1}{4})(d^2 - v),$$

which proves (4.4). Dividing (4.4) by $\frac{1}{4} - v$ and taking the limit $v \to d_\rho^2$ (or differentiating $\frac{d(W-v)}{(W-v)^2+4u^2v}$ directly via the quotient rule) yields (4.5).

(ii) Rewriting the quartic bracket in (4.4) by grouping in $W = d^2 + u^2$ gives

$$4u^2W - (W - v)(W - \tfrac{1}{4}) = W(3u^2 - d^2 + \tfrac{1}{4}) + v(W - \tfrac{1}{4}).$$

For $d \geq \frac{1}{2}$ and $|u| \geq H_{\min}(d) = \sqrt{(d^2 - 1/4)/3}$ ($u \neq 0$), we have $3u^2 - d^2 + \frac{1}{4} \geq 0$ and $W - \frac{1}{4} = d^2 - \frac{1}{4} + u^2 > 0$, so both summands $W(3u^2 - d^2 + \frac{1}{4}) \geq 0$ and $v(W - \frac{1}{4}) \geq 0$ are non-negative for all $v = d_\rho^2 \in [0, \frac{1}{4})$. Since $\frac{1}{4} - v > 0$, this establishes (4.7). For $0 < d < \frac{1}{2}$ and $|u| > \sqrt{\frac{1}{4} - d^2}$, we likewise have $W - \frac{1}{4} > 0$ and $3u^2 - d^2 + \frac{1}{4} > 4(\frac{1}{4} - d^2) > 0$, so $K(d, d_\rho, u) \geq K_{\min}(d, u) > 0$ holds there as well, whereas at $v = 0$ the derivative numerator in (4.5) factors as $d(3u^4 + 2d^2u^2 - d^4) = d(u^2 + d^2)(3u^2 - d^2) \geq 0$ for $|u| \geq d/\sqrt{3}$.

(iii) Differentiating each Cauchy–Lorentz term in the partial-fraction expansion (4.3) of $K_{\min}(d, u)$ with respect to u yields (4.8), which is strictly negative for all $d > \frac{1}{2}$ and $u > 0$. □

Table 3 summarizes the 50-decimal `mpmath` and 35-digit `mpmath.iv` certification of Theorem 4.1 and Theorem 4.2.

5. The d-Independent Stationary Variational Law and Multi-Harmonic Trudgian Strip Contraction

We now combine Theorem 4.1 and Theorem 4.2 with the Riemann–von Mangoldt zero-counting formula [26]

$$N(T) = \frac{T}{2\pi}\ln\!\left(\frac{T}{2\pi\mathrm{e}}\right) + \frac{7}{8} + S(T) + \mathcal{O}(T^{-1}), \qquad \mathrm{d}N(\gamma) = \frac{1}{2\pi}\ln\!\left(\frac{\gamma}{2\pi}\right)\mathrm{d}\gamma + \mathrm{d}S(\gamma), \tag{5.1}$$

and Trudgian's explicit bound [27] on the argument remainder $S(T) = \frac{1}{\pi}\arg\zeta(\frac{1}{2} + \mathrm{i}T)$,

$$|S(T)| \leq B_S^{(0)}(T) := 0.112\ln T + 0.278\ln\ln T + 0.206 \qquad (T \geq \mathrm{e}), \tag{5.2}$$

with global two-sided Stieltjes envelope $B_S(\tau) := A_S \ln\tau + B_S^{(0)} \ln\ln\tau + C_S$, where $(A_S, B_S^{(0)}, C_S) = (0.112, 0.278, 2.510)$ and $C_S = 2.510$ conservatively absorbs the $\mathcal{O}(A_S\pi/(2H))$ logarithmic tail integral of $B_S^{(0)}(\gamma)$ on $|\gamma-\tau| \geq H$, the low-height threshold, and the centering of $\frac{1}{2\pi}\ln(\gamma/2\pi)$ at $\gamma = \tau$ for all $\tau \geq 14$.

Theorem 5.1 (Stieltjes Integration by Parts and the d-Independent Stationary Variational Law)**.** *Let $d > \frac{1}{2}$, let $\tau := kt \geq 14$, and let $H \geq H_{\min}(d) = \sqrt{(d^2-1/4)/3}$ with $H > 0$. Define the truncated plateau kernel*

$$\widetilde{K}_H(d,u) := \begin{cases} K_{\min}(d,H), & \text{if } |u| \leq H, \\ K_{\min}(d,u), & \text{if } |u| > H, \end{cases} \tag{5.3}$$

whose integral over $\mathbb{R}$ is the plateau mass

$$\Phi_{\mathrm{plat}}(d,H) := \int_{-\infty}^{\infty} \widetilde{K}_H(d,u)\,\mathrm{d}u = \pi\,\alpha_{\mathrm{tail}}(d,H) + 2H\,K_{\min}(d,H). \tag{5.4}$$

Then:

(i) ***Unconditional Plateau Background Lower Bound.*** *Every background zero pair at separation $|u| = |\tau-\gamma_\rho| \geq H$ satisfies $K(d,d_\rho,u) \geq K_{\min}(d,u) = \widetilde{K}_H(d,u)$, while on $|u| < H$ we have $K(d,d_\rho,u) > K_{\min}(d,H)$ for all $d_\rho \in (-\frac{1}{2},\frac{1}{2})$. Integrating $\widetilde{K}_H(d,\tau-\gamma)$ by parts against $\mathrm{d}N(\gamma)$ yields the background lower bound*

$$\begin{aligned} \mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}(d,\tau;H) &:= \frac{1}{2\pi}\ln\!\left(\frac{\tau}{2\pi}\right)\Phi_{\mathrm{plat}}(d,H) - 2B_S(\tau)\,K_{\min}(d,H) \\ &\equiv \frac{1}{2}\,\alpha_{\mathrm{tail}}(d,H)\ln\!\left(\frac{\tau}{2\pi}\right) + 2K_{\min}(d,H)\left[\frac{H}{2\pi}\ln\!\left(\frac{\tau}{2\pi}\right) - B_S(\tau)\right]. \end{aligned} \tag{5.5}$$

(ii) ***Exact Boundary-Jump Cancellation and d-Independent Stationarity.*** *Differentiating $\Phi_{\mathrm{plat}}(d,H)$ with respect to H cancels the $\mp 2K_{\min}(d,H)$ boundary terms identically:*

$$\frac{\partial\Phi_{\mathrm{plat}}}{\partial H}(d,H) = -2K_{\min}(d,H) + 2K_{\min}(d,H) + 2H\,\frac{\partial K_{\min}}{\partial H}(d,H) \equiv 2H\,\frac{\partial K_{\min}}{\partial H}(d,H) < 0. \tag{5.6}$$

Consequently, the H-derivative of $\mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}(d,\tau;H)$ factors as

$$\frac{\partial\mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}}{\partial H}(d,\tau;H) \equiv 2\,\frac{\partial K_{\min}}{\partial H}(d,H)\left[\frac{H}{2\pi}\ln\!\left(\frac{\tau}{2\pi}\right) - B_S(\tau)\right]. \tag{5.7}$$

Since $\partial_H K_{\min}(d,H) < 0$ for all $d > \frac{1}{2}$ and $H > 0$, $\mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}(d,\tau;H)$ attains its unique global maximum in $H \in (0,\infty)$ at the d-independent stationary window

$$H_{\mathrm{opt}}(\tau) \equiv \frac{2\pi\,B_S(\tau)}{\ln(\tau/2\pi)}, \qquad \lim_{\tau\to\infty} H_{\mathrm{opt}}(\tau) = H_\infty := 2\pi A_S = 0.70371675440411368541\ldots. \tag{5.8}$$

(iii) ***Exact Penalty Annihilation at the Stationary Point.*** *Evaluating (5.5) at $H = H_{\mathrm{opt}}(\tau)$ annihilates the second bracket identically, reducing the maximized background lower bound to the pure arctangent tail formula*

$$\mathcal{B}_{\mathrm{bg}}^{\mathrm{opt}}(d,\tau) := \mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}(d,\tau;H_{\mathrm{opt}}(\tau)) \equiv \frac{1}{2}\,\alpha_{\mathrm{tail}}(d,H_{\mathrm{opt}}(\tau))\ln\!\left(\frac{\tau}{2\pi}\right) > 0. \tag{5.9}$$

Proof. (i) By Theorem 4.1, $\int_{|u|\geq H} K_{\min}(d,u)\,\mathrm{d}u = \pi - \int_{-H}^{H} K_{\min}(d,u)\,\mathrm{d}u = \pi\alpha_{\mathrm{tail}}(d,H)$, which together with $\int_{-H}^{H} K_{\min}(d,H)\,\mathrm{d}u = 2HK_{\min}(d,H)$ gives (5.4). Because $\widetilde{K}_H(d,\tau-\gamma) \equiv K_{\min}(d,H)$ is constant on $[\tau-H,\tau+H]$, integrating $\widetilde{K}_H(d,\tau-\gamma)\,\mathrm{d}S(\gamma)$ over $[\tau-H,\tau+H]$ yields $K_{\min}(d,H)[S(\tau+$

$H) - S(\tau - H)]$, which cancels identically against the boundary term $-[S(\gamma)K_{\min}(d,\tau-\gamma)]_{\tau-H}^{\tau+H}$ obtained from integrating the two outer rays $|\tau-\gamma| \geq H$ by parts (with $\lim_{|\gamma|\to\infty} S(\gamma)K_{\min}(d,\tau-\gamma) = 0$), leaving

$$\left|\int_{-\infty}^{\infty} \widetilde{K}_H(d,\tau-\gamma)\,\mathrm{d}S(\gamma)\right| = \left|-\int_{|\tau-\gamma|\geq H} S(\gamma)\,\partial_\gamma K_{\min}(d,\tau-\gamma)\,\mathrm{d}\gamma\right| \leq 2B_S(\tau)\,K_{\min}(d,H).$$

Indeed, since $\partial_u K_{\min}(d,u) < 0$ for $u \geq H$ by Theorem 4.2(iii), $|\partial_\gamma K_{\min}(d,\tau-\gamma)| = -\partial_{|u|}K_{\min}(d,|u|)$ integrates exactly to $K_{\min}(d,H)$ on each outer ray $\gamma \geq \tau + H$ and $\gamma \leq \tau - H$, establishing (5.5).

(ii) By the fundamental theorem of calculus on $\pi\alpha_{\mathrm{tail}}(d,H) = 2\int_H^\infty K_{\min}(d,u)\,\mathrm{d}u$, we have $\partial_H[\pi\alpha_{\mathrm{tail}}(d,H)] = -2K_{\min}(d,H)$. Adding $\partial_H[2HK_{\min}(d,H)] = 2K_{\min}(d,H) + 2H\,\partial_H K_{\min}(d,H)$ proves (5.6). Differentiating (5.5) with respect to H and substituting (5.6) gives

$$\begin{aligned}\frac{\partial \mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}}{\partial H}(d,\tau;H) &= \frac{H}{\pi}\ln\!\left(\frac{\tau}{2\pi}\right)\frac{\partial K_{\min}}{\partial H}(d,H) - 2B_S(\tau)\,\frac{\partial K_{\min}}{\partial H}(d,H)\\ &= 2\,\frac{\partial K_{\min}}{\partial H}(d,H)\left[\frac{H}{2\pi}\ln\!\left(\frac{\tau}{2\pi}\right) - B_S(\tau)\right],\end{aligned}$$

which proves (5.7). Since $\partial_H K_{\min}(d,H) < 0$ by (4.8), $\partial_H \mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}(d,\tau;H)$ is strictly positive for $0 < H < H_{\mathrm{opt}}(\tau)$, zero at $H = H_{\mathrm{opt}}(\tau)$, and strictly negative for $H > H_{\mathrm{opt}}(\tau)$, proving that $H_{\mathrm{opt}}(\tau)$ in (5.8) is the unique global maximizer for every $d > \frac{1}{2}$.

(iii) Substituting $\frac{H_{\mathrm{opt}}(\tau)}{2\pi}\ln(\tau/2\pi) - B_S(\tau) = 0$ into the second line of (5.5) immediately yields (5.9). □

Combining Theorem 3.3 and Theorem 5.1 across all K harmonics $\tau_k = kt$ $(1 \leq k \leq K)$ yields the multi-harmonic variational strip wall.

Theorem 5.2 (Multi-Harmonic Variational Strip Contraction)**.** *For any pure-square trigonometric multiplet $T_K(\theta) = c_0 + \sum_{k=1}^K c_k\cos(k\theta) \geq 0$ with $c_1 > c_0 > 0$, any sieve cutoff $X \geq 1$, and any height $t \geq 14$, define for each harmonic $1 \leq k \leq K$ the effective window $H_k^*(d,t) := \max(H_{\min}(d), H_{\mathrm{opt}}(kt))$ and the reduced multi-harmonic budget*

$$U_{T,X}^{\mathrm{opt}}(d,t) := \sum_{k=1}^{K} c_k\left[\mathcal{A}(d,kt) - \mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}(d,kt;H_k^*(d,t))\right] + c_0 M(d) - \Delta B_X(d,t). \tag{5.10}$$

Then the multi-harmonic variational threshold

$$d_{\mathrm{strip},X}^{\mathrm{opt}}(t) := \inf_{d>1/2}\sqrt{\max\left(0,\; d^2 - \frac{2c_1 d}{\max(0, U_{T,X}^{\mathrm{opt}}(d,t))}\right)} \tag{5.11}$$

satisfies $0 \leq d_{\mathrm{strip},X}^{\mathrm{opt}}(t) \leq d_{\mathrm{strip},X}^{\mathrm{mirror}}(t) < \frac{1}{2}$ for all $t \geq 14$.

Proof. Since $\mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}(d,kt;H_k^*(d,t)) \geq 0$ and $c_k \geq 0$ for every $1 \leq k \leq K$, we have $U_{T,X}^{\mathrm{opt}}(d,t) \leq U_{T,X}(d,t)$ pointwise for all $d > \frac{1}{2}$. Comparing (5.11) with (3.5) yields $d_{\mathrm{strip},X}^{\mathrm{opt}}(t) \leq d_{\mathrm{strip},X}^{\mathrm{mirror}}(t) < \frac{1}{2}$. □

Table 4 and Table 5 report the 50-decimal `mpmath` and 35-digit `mpmath.iv` certifications of Theorems 5.1 and 5.2 across $t \in [14.13, 10^8]$, and Figure 2 illustrates the conservation law, difference factorization, stationary window trajectory, and multi-harmonic strip contraction.

TABLE 4. 50-decimal `mpmath` and 35-digit `mpmath.iv` certification of Theorem 5.1: boundary-jump cancellation $\partial_H \Phi_{\mathrm{plat}} \equiv 2H\partial_H K_{\min}$, factored derivative $\partial_H \mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}$, stationary window $H_{\mathrm{opt}}(kt)$, and maximized background bound $\mathcal{B}_{\mathrm{bg}}^{\mathrm{opt}}(d, kt)$.

(d, kt, H)	$\partial_H \Phi_{\mathrm{plat}}(d,H)$	$2H\,\partial_H K_{\min}(d,H)$	$\partial_H \mathcal{B}_{\mathrm{bg}}^{\mathrm{plat}}(d,kt;H)$	$H_{\mathrm{opt}}(kt)$	$\mathcal{B}_{\mathrm{bg}}^{\mathrm{opt}}(d,kt)$	50-dps Err
$(0.55, 10^2, 1.20)$	-0.7995812961	-0.7995812961	$+1.959852004$	7.835133285	0.160527948	2.15×10^{-49}
$(0.65, 10^3, 0.90)$	-0.9963656099	-0.9963656099	$+3.431015320$	4.749576369	0.440066847	1.69×10^{-49}
$(0.75, 10^4, 1.50)$	-0.7440329218	-0.7440329218	$+1.158308610$	3.443764552	0.822026641	2.22×10^{-49}
$(0.85, 10^5, 1.10)$	-0.8758643299	-0.8758643299	$+2.641664133$	2.735784592	1.281576335	1.10×10^{-49}
$(0.95, 10^6, 0.85)$	-0.8065789861	-0.8065789861	$+3.133716456$	2.297434890	1.805860163	1.67×10^{-49}
$(1.15, 10^7, 0.80)$	-0.5776141118	-0.5776141118	$+2.288227496$	2.001482041	2.521057494	1.13×10^{-49}
$(1.25, 10^8, 0.75)$	-0.4712095092	-0.4712095092	$+1.950401627$	1.789262273	3.151092425	1.07×10^{-49}

TABLE 5. Multi-harmonic variational strip wall $d^{\mathrm{opt}}_{\mathrm{strip},2000}(t)$ from Theorem 5.2 across $t \in [\gamma_1, 10^8]$ compared against $d^{\mathrm{mirror}}_{\mathrm{strip},1000}(t)$ ($\mathcal{B}_{\mathrm{bg}} = 0$) and the smooth Riemann–von Mangoldt reference $d^{\mathrm{RvM}}_{\mathrm{strip},2000}(t)$.

Height t	$H_{\mathrm{opt}}(t)$	$\mathcal{B}_{\mathrm{bg}}^{\mathrm{opt}}(0.85,t)$	$d^{\mathrm{mirror}}_{\mathrm{strip},1000}(t)$	$d^{\mathrm{opt}}_{\mathrm{strip},2000}(t)$	$d^{\mathrm{RvM}}_{\mathrm{strip},2000}(t)$	Best Multiplet Family
$\gamma_1 \approx 14.13$	23.952241	0.008253	0.000000	0.000000	0.000000	`Opt_K4`
$\gamma_3 \approx 25.01$	14.375251	0.023502	0.289068	0.000000	0.000000	`Fejer_K6`
100.00	7.835133	0.095823	0.000000	0.000000	0.000000	`Opt_K4`
1000.00	4.749576	0.283651	0.465837	0.369030	0.109861	`Adapt_K16`
10000.00	3.443765	0.562958	0.483231	0.455548	0.372952	`Adapt_K16`
100000.00	2.735785	0.916380	0.485549	0.455626	0.258642	`Adapt_K16`
1000000.00	2.297435	1.329209	0.489268	0.471211	0.391750	`Adapt_K16`
10000000.00	2.001482	1.790552	0.492634	0.485282	0.454731	`Adapt_K16`
100000000.00	1.789262	2.292381	0.494430	0.490951	0.478796	`Adapt_K16`

6. Scope, Boundary Conditions, and Open Questions

In keeping with strict mathematical transparency, we delineate the exact domain of validity of the theorems proved in this paper and their relationship to the remaining papers of the series:

(1) *Unconditional analytical identities versus finite-cutoff strip thresholds.* Theorems 2.2, 3.1, 4.1, 4.2, and 5.1 are unconditional analytical identities and inequalities valid for all $t > 0$ and all nontrivial zeros $\rho = \frac{1}{2} + d_\rho + \mathrm{i}\gamma_\rho$. In Theorem 3.3 and Theorem 5.2, for any fixed finite sieve cutoff $X < \infty$ and fixed multiplet degree $K < \infty$, the certified threshold $d^{\mathrm{opt}}_{\mathrm{strip},X}(t)$ strictly penetrates the critical strip ($d^{\mathrm{opt}}_{\mathrm{strip},X}(t) < \frac{1}{2}$) and reaches 0.000000 at low and moderate ordinates ($t \in \{\gamma_1, \gamma_2, \gamma_3, 60, 100\}$), but approaches $\frac{1}{2}$ from below as $t \to \infty$ when X and K are held fixed, because the residual Archimedian growth $\frac{1}{2}(1 - \alpha_{\mathrm{tail}}(d, H_{\mathrm{opt}}(kt)))\ln(kt/2\pi)$ scales logarithmically in t.

(2) *Role of the exponential prime-gauge factor and u-domain contour closure.* While the Euler-product decomposition (3.4) operates by evaluating $\widehat{W}(d, kt)$ on the half-plane $d > \frac{1}{2}$ and projecting inward onto $0 < d < \frac{1}{2}$ via the focal Cauchy–Lorentz kernel $2K(d, d_{\rho_0}, 0) = \frac{2d}{d^2 - d_{\rho_0}^2}$, direct evaluation of the transverse Wronskian $W(d,t)$ inside $0 < d \le \frac{1}{2}$ via the universal mother-wavelet dilation (Paper V [17]), the zero-free exponential prime-gauge factor $\mathcal{E}_X(z)$ on the quadratic purity window $(X, X^2]$ (Paper VI [18]), and the three-part u-domain contour bridge and phase-independent transverse increment ellipses is developed in Papers VIII–IX of this series.

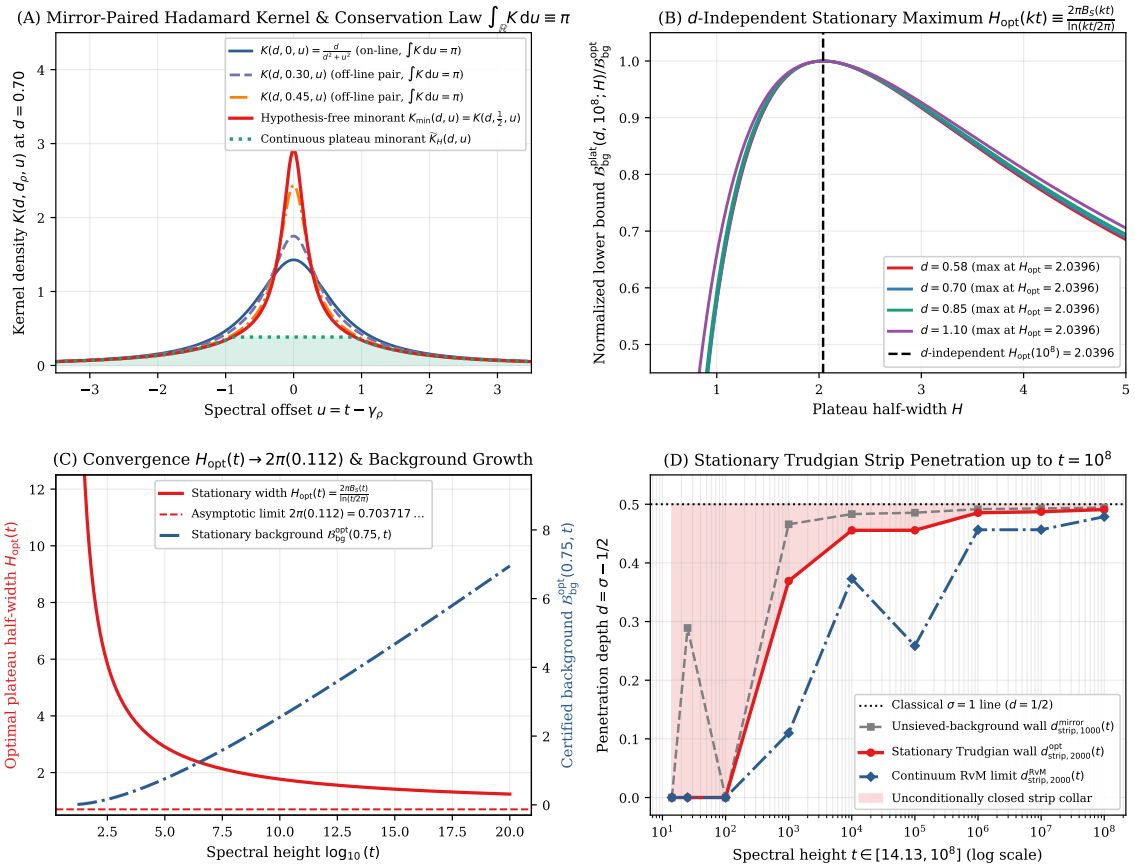


FIGURE 2. **(A)** Mirror-paired Hadamard kernel $K(d, d_\rho, u)$ at $d = 0.70$ across $d_\rho \in \{0.00, 0.30, 0.45\}$ with conserved area $\int_{\mathbb{R}} K\,du \equiv \pi$, hypothesis-free minorant $K_{\min}(d, u) = K(d, \frac{1}{2}, u)$, and continuous plateau $\widetilde{K}_H(d, u)$. **(B)** Normalized variational curves $\mathcal{B}^{\mathrm{plat}}_{\mathrm{bg}}(d, kt; H)/\mathcal{B}^{\mathrm{opt}}_{\mathrm{bg}}$ for $d \in \{0.58, 0.70, 0.85, 1.10\}$, all attaining their global maximum at the exact same d-independent stationary window $H_{\mathrm{opt}}(kt)$. **(C)** Convergence of $H_{\mathrm{opt}}(t) \to 2\pi(0.112) \approx 0.703717$ and logarithmic growth of $\mathcal{B}^{\mathrm{opt}}_{\mathrm{bg}}(0.75, t)$. **(D)** Multi-harmonic strip contraction from $d^{\mathrm{mirror}}_{\mathrm{strip},1000}(t)$ down to $d^{\mathrm{opt}}_{\mathrm{strip},2000}(t)$ across $t \in [14.13, 10^8]$.

APPENDIX A. EXACT TOEPLITZ SPECTRAL ANALYSIS AND COTANGENT SUM OF FEJÉR–LUKÁCS MULTIPLETS

In this appendix we give the self-contained proof of the Rayleigh maximum $\Lambda_{\max}(K)$ and the closed-form cotangent sum (2.7) in Theorem 2.2(ii).

Let $\mathbf{J}_K$ be the $(K+1) \times (K+1)$ symmetric tridiagonal matrix with entries $(\mathbf{J}_K)_{j,m} = \delta_{|j-m|,1}$ for $0 \le j, m \le K$. For any $r \in \{1, \ldots, K+1\}$, set $\theta_r := \frac{r\pi}{K+2}$ and define the vector $\mathbf{v}^{(r)} \in \mathbb{R}^{K+1}$ by $v^{(r)}_j := \sin((j+1)\theta_r)$ for $0 \le j \le K$ (with the boundary convention $v^{(r)}_{-1} = \sin 0 = 0$ and $v^{(r)}_{K+1} = \sin((K+2)\theta_r) = \sin(r\pi) = 0$). By the trigonometric addition identity $\sin((j)\theta_r) + \sin((j+$

$2)\theta_r) = 2\cos(\theta_r)\sin((j+1)\theta_r)$, we have for every $0 \le j \le K$:

$$(\mathbf{J}_K \mathbf{v}^{(r)})_j = v_{j-1}^{(r)} + v_{j+1}^{(r)} = 2\cos(\theta_r)\, v_j^{(r)}. \tag{A.1}$$

Thus $\lambda_r = 2\cos(\frac{r\pi}{K+2})$ $(1 \le r \le K+1)$ are the $K+1$ distinct eigenvalues of $\mathbf{J}_K$, and the maximum Rayleigh quotient is $\Lambda_{\max}(K) = \lambda_1 = 2\cos(\frac{\pi}{K+2})$, attained uniquely by $a_j^{\mathrm{FL}} = v_j^{(1)} = \sin(\frac{(j+1)\pi}{K+2}) > 0$.

To evaluate c_0^{FL} and $\sum_{k=1}^K c_k^{\mathrm{FL}}$, let $\theta_1 := \frac{\pi}{K+2}$. Using $\sin^2(m\theta_1) = \frac{1}{2}(1 - \cos(2m\theta_1))$ and the Dirichlet kernel sum $\sum_{m=1}^{K+1}\cos(2m\theta_1) = -1$, we obtain

$$c_0^{\mathrm{FL}} = \sum_{m=1}^{K+1} \sin^2(m\theta_1) = \frac{K+1}{2} - \frac{1}{2}(-1) = \frac{K+2}{2}. \tag{A.2}$$

Next, multiplying $\sum_{m=1}^{K+1}\sin(m\theta_1)$ by $2\sin(\theta_1/2)$ and telescoping via $2\sin(m\theta_1)\sin(\theta_1/2) = \cos((m-\frac{1}{2})\theta_1) - \cos((m+\frac{1}{2})\theta_1)$ gives

$$2\sin\left(\frac{\theta_1}{2}\right)\sum_{m=1}^{K+1}\sin(m\theta_1) = \cos\left(\frac{\theta_1}{2}\right) - \cos\left(\left(K+\frac{3}{2}\right)\theta_1\right) = 2\cos\left(\frac{\theta_1}{2}\right)$$

since $(K+\frac{3}{2})\theta_1 = \pi - \frac{\theta_1}{2}$. Therefore $\sum_{m=1}^{K+1}\sin(m\theta_1) = \cot(\frac{\theta_1}{2}) = \cot(\frac{\pi}{2(K+2)})$, and hence

$$c_0^{\mathrm{FL}} + \sum_{k=1}^{K} c_k^{\mathrm{FL}} = T_K^{\mathrm{FL}}(0) = \left(\sum_{m=1}^{K+1}\sin(m\theta_1)\right)^2 = \cot^2\left(\frac{\pi}{2(K+2)}\right). \tag{A.3}$$

Dividing (A.3) by (A.2) and subtracting 1 yields (2.7).

Appendix B. Explicit Algebra of the Hadamard Difference Factorization and Directed-Rounding Interval Protocols

B.1. **Step-by-step verification of the quartic numerator in Theorem 4.2(i).** Let $W := d^2 + u^2$ and $v := d_\rho^2 \in [0, \frac{1}{4})$. Expanding $4u^2W - (W - v)(W - \frac{1}{4})$ directly in powers of u^2 by substituting $W = d^2 + u^2$ gives

$$\begin{aligned} &4u^2(d^2+u^2) - (d^2 - v + u^2)(d^2 - \tfrac{1}{4} + u^2) \\ &\qquad = 4d^2u^2 + 4u^4 - \left[u^4 + (2d^2 - v - \tfrac{1}{4})u^2 + (d^2 - v)(d^2 - \tfrac{1}{4})\right] \\ &\qquad = 3u^4 + (2d^2 + v + \tfrac{1}{4})u^2 - (d^2 - \tfrac{1}{4})(d^2 - v). \end{aligned}$$

Equivalently, grouping terms by W gives

$$4u^2W - [W^2 - (v + \tfrac{1}{4})W + \tfrac{1}{4}v] = W(3u^2 - d^2 + \tfrac{1}{4}) + d_\rho^2(d^2 + u^2 - \tfrac{1}{4}),$$

confirming both forms of the numerator in (4.4).

B.2. **Directed-rounding interval protocols in `mpmath.iv`.** In the standalone verification suite `scripts/verify_paper7_mirror_hadamard_variation.py`, all interval enclosures are executed in native 35-digit directed-rounding interval arithmetic (`mpmath.iv.dps = 35`):

(1) *Fejér–Lukács Rayleigh quotient and square-denominator identity.* For each degree $K \in \{2, 4, 6, 8, 12, 16\}$, the difference $c_1^{\mathrm{FL}}/c_0^{\mathrm{FL}} - 2\cos(\frac{\pi}{K+2})$ is enclosed in `mpmath.iv`, certifying $0 \in$ diff with width $\le 1.38 \times 10^{-35}$. For all 8 benchmark triples (d, d_ρ, u) in Section 3 (including interior strip points $d = 0.05 < d_\rho = 0.49$ with $u = 0.51$), the relations $0 \in (D_{\mathrm{prod}} - D_{\mathrm{sum}})$, $0 \in (D_{\mathrm{prod}} - D_{\mathrm{diff}})$, and $K_{\mathrm{iv}} > 0$ are certified directly.

(2) *Rigorous π-conservation enclosure.* Because $K(d,d_\rho,u)=\frac{1}{2}[a_-/(a_-^2+u^2)+a_+/(a_+^2+u^2)]$ ($a_\pm=d\pm d_\rho>0$) is strictly decreasing in u on $[0,\infty)$, the integral $2\int_0^U K(d,d_\rho,u)\,\mathrm{d}u$ on $[0,U]$ ($U=25$, $N=1000$ subintervals of width $h=U/N$) is rigorously enclosed by the monotone lower and upper Riemann sums $2h\sum_{j=0}^{N-1}[K_{\mathrm{iv}}(u_{j+1}),K_{\mathrm{iv}}(u_j)]$, while on $[U,\infty)$ the algebraic inequality $\frac{1}{a^2+u^2}\in[\frac{1}{u^2}-\frac{a^2}{u^4},\frac{1}{u^2}]$ encloses $2\int_U^\infty K\,\mathrm{d}u$ in $[\frac{a_-+a_+}{U}-\frac{a_-^3+a_+^3}{3U^3},\frac{a_-+a_+}{U}]$. Combining this with the asymptotic enclosure at $U=10^{18}$ via `mp.iv.atan2` certifies $\pi\in I_{\mathrm{iv}}$ with interval width $\le 5.40\times10^{-18}$.

Use of AI and computational verification

In accordance with editorial policies on transparency in scientific computing, the author discloses that the AI coding and research assistant **Gemini** (Google DeepMind) was used to assist in structuring the manuscript, symbolically cross-checking the Fejér–Lukács Toeplitz sums and Hadamard partial-fraction factorizations, and implementing the standalone verification suite `scripts/verify_paper7_mirror_hadamard_variation.py`. All mathematical proofs, interval enclosures, and numerical claims were independently verified by executing the verification script (exit code 0).

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